Integration — Lesson 14 Test: The Average Value of a Function

10 questions · show your work in the box · calculators allowed · remember: average = total ÷ length of interval

Name: ______________________________ Date: ______________ Score: ______ / 10

1. Find the average value of the constant function $f(x) = 9$ on the interval $[0, 5]$. (Predict first, then show the formula agrees.) 1 pt

2. Find the average value of $f(x) = 3x$ on $[0, 4]$. Check your answer by averaging the two endpoint values (allowed here — say why). 1 pt

3. Find the average value of $f(x) = x^2$ on $[0, 12]$. 1 pt

4. The average value of f on [0, 6] is 8. What is $\int_0^6 f(x)\,dx$?

1 pt

5. Find the average value of $f(x) = x + 5$ on [0, 8].

1 pt

6. Starting at $t = 0$ (in hours), the temperature in a garage is $T(t) = 60 + 2t$ degrees Fahrenheit. What is the average temperature from $t = 0$ to $t = 10$?

1 pt

7. A remote-control car's velocity is $v(t) = 6t^2$ m/s from $t = 0$ to $t = 2$ seconds. Find (a) the total distance it travels and (b) its average velocity.

1 pt

8. The average value of $f(x) = x^2$ on the interval $[0, c]$ is 3. Find c. (Hint: first write the average in terms of c.) 1 pt

9. A classmate claims the average of $f(x) = x^2$ on $[0, 6]$ is $(f(0) + f(6))/2 = 18$. The true average is 12. Explain in a sentence or two why the endpoint method overshoots for x^2, and name the kind of function it DOES work for. 1 pt

10. A particle's velocity is $v(t) = t^2 + 2t$ m/s from $t = 0$ to $t = 3$ seconds. (a) Find the net change in its position. (b) Find its average velocity. (c) Check that the constant average velocity for 3 seconds gives the same net change. 1 pt